MIL-K-8144B(USAF)
19 Oct 83
SUPERSEDING
MIL-K-8144A(USAF)
3 Oct 1968

MILITARY SPECIFICATION

KEYER, OVERHEAT WARNING LIGHT, TYPE MA-1

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. Scope

1.1 Scope. This specification covers one type of overheating warning light keyer, designated Type MA-1.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

PPP-B-601	Boxes, Wood, Cleated - Plywood
PPP-B-636	Boxes, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Method of
DOD-D-1000	Drawings, Engineering and Associated Lists

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Table for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6340

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MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MS25237	Lamp, Incandescent, T-1 3/4, Bulb, Midget, Flange Base

DRAWINGS

AIR FORCE

68C37331	Keyer Overheat Warning Light

* (Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951	Standard Practice for Commercial Packaging

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

* 2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

* 3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

* 3.1.1 Each unit shipped shall be certified as meeting the design requirements of Air Force Drawing 68C37331.

* 3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

* 3.3 Material.

* 3.3.1 Standard parts. AN or MIL Standard parts shall be used wherever they are suitable for the purpose and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, et cetera, may be used, provided they have suitable properties and are replaceable by the AN or MIL Standard parts without alteration, and provided the corresponding AN or MIL Standard part numbers are referenced on the drawings and in the parts lists. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

* 3.3.2 Protective treatment. When materials are used in the construction of the keyer that are subject to deterioration when exposed to climatic and

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environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

* 3.3.3 Metals. The use of dissimilar metals in contact which tend toward active electrolytic corrosion is not acceptable, however, metal plating or spraying of dissimilar base metals to provide similar or suitable abutting surfaces is permitted. The term "dissimilar metals" as used in this specification is defined in MIL-STD-889.

* 3.4 Design and construction. The keyer shall be of compact design and of sufficiently rugged construction to withstand the mechanical and thermal shocks and stresses incident to shipping, storage, installation, and its use in aircraft.

* 3.4.1 Operating frequency. When connected to a 28 volt supply, the keyer shall be capable of making and breaking the current of an MS25273-327 (red) lamp (minimum) at a rate of 100 ±10 cycles per minute (CPM) at an ambient temperature of 21° Centigrade (C).

* 3.4.1.1 Voltage frequency. The keyer shall operate at a frequency of 100 ±30 cpm for any applied voltage between 22 and 32 volts direct current (DC) at an ambient temperature of 21°C. A 50/50 duty cycle on the "ON" and "OFF" ratio shall be maintained.

* 3.4.2 The maximum dimensions of the keyer shall be 2.15 inches in length by 1.4 inches in width and 1.5 inches in height, excluding mounting flange, terminal and terminal cover.

* 3.4.3 The total weight of the keyer and its mount shall not exceed .45 pounds.

* 3.4.4 The keyer shall have a maximum starting time of one second after application of power.

* 3.4.5 Life. The keyer shall be capable of a minimum of 500,000 cycles of operation with an applied voltage of 28 volts DC at an ambient temperature of 21°C.

* 3.4.6 The keyer shall be designed so that it meets the radio interference requirements of MIL-STD-461.

* 3.4.7 The internal voltage drop of the keyer shall not exceed 1.5 volts DC.

* 3.5 Performance. The keyer shall be capable of operating under the following conditions:

* 3.5.1 Temperature. A temperature range of -54 to 80°C.

* 3.5.2 Altitude. A pressure altitude range from 0 to 50,000 feet.

* 3.5.3 Humidity. A relative humidity of 95 +5-3 percent at a temperature of 65°C.

* 3.5.4 Sand and dust. Exposure to airborne sand and dust particles encountered in normal and desert operation.

* 3.5.5 Salt spray. Exposure to atmosphere containing salt-laden moisture.

* 3.5.6 Fungi. Moist fungi encountered in tropical and semitropical climates.

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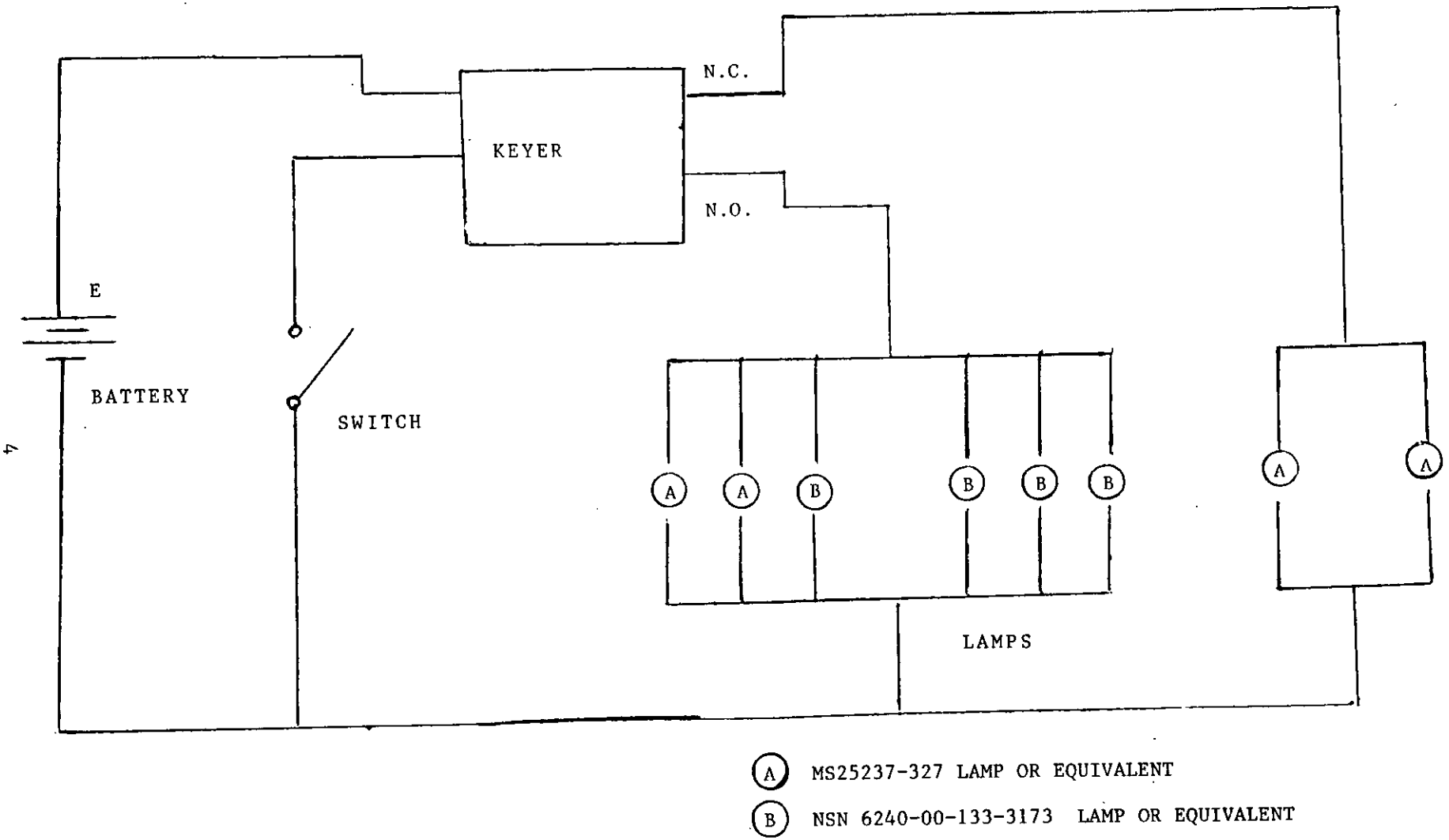


FIGURE 1. Keyer circuit.

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* 3.5.7 Vibration. Vibration incident to use on aircraft.

* 3.5.8 The frequency of operation shall not deviate more than ±20 percent as a result of operation under any of the above conditions, except for temperature extremes.

* 3.5.9 Voltage Surges. Components of the warning light keyer shall be protected against high-voltage spikes of 200 volts DC, of 1 microsecond time duration. (Transient as specified in MIL-STD-704).

* 3.5.9.1 Input transient voltage protection. An additional design requirement shall be that a signal applied to the Normal Open (NO) terminal shall not be present at any other terminal under any condition.

* 3.5.10 Load rating. Keyer shall be capable of operation with loads of 40MA to 1AMP in each leg.

* 3.6 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the requirements of DOD-D-1000.

* 3.7 Identification of product. Each keyer shall be marked for identification in accordance with MIL-STD-130. The nameplate shall contain the information presented below:

Keyer, Overheat Warning Light
Type MA-1
Specification MIL-K-8144
Manufacturer's Part Number _____
(Manufacturer's Name or Trademark)
U.S. Property, FSN 6340-674-2289

* 3.8 The electrical rating of the keyer shall be permanently and legibly marked on the exterior surface.

* 3.9 Workmanship.

* 3.9.1 General. The keyer shall be fabricated and finished in accordance with high grade aircraft manufacturing practice covering this class of equipment. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions and marking of parts.

* 3.9.2 The keyer shall be throughly cleaned, and metal chips and other foreign material shall be removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.2 Classification of tests. The inspection and testing of the keyer shall be classified as follows:

a. First article tests (see 4.4).

b. Quality conformance inspection (see 4.5).

* 4.3 Test conditions.

* 4.3.1 Prior to testing, the keyer shall be connected as shown in Figure 1.

* 4.4 First article tests.

* 4.4.1 First article test samples. Six keyers representative of the production keyer shall be furnished for preproduction testing. Each sample shall be subjected to the examination of product (see 4.5.2.1) and all the tests specified under 4.4.2 except the life test (see 4.4.2.3) and the environmental tests (see 4.4.2.5). The keyers shall then be divided into two groups of three each. Group one shall be subjected to the life tests and group two shall be subjected to the environmental tests.

* 4.4.2 The first article tests shall consist of the following tests:

* 4.4.2.1 Operating frequency test. Using the circuit shown on Figure 1, all samples shall be tested to determine their operating frequency. The frequency shall be within the limits specified in 3.4.1 with an applied voltage of 28 volts.

* 4.4.2.2 Voltage frequency test. The frequency shall be within the limits specified in 3.4.1.1 for any applied voltage between 22 and 32 volts DC.

* 4.4.2.3 Life test. The samples shall be connected to the circuit shown on Figure 1 and 28 volts applied for a period of 1 minute. At the end of one minute the circuit shall be opened for a period of 30 seconds. This cycle shall be repeated for a total of 1000 cycles. At the end of 1000 cycles, the keyer shall be operated continuously for 1 hour. At the end of the test, the operating frequency of the keyer shall be within ±20 cpm of its frequency before the test.

* 4.4.2.4 Radio interference test. The keyer shall be tested to determine compliance with the requirements of MIL-STD-461.

* 4.4.2.5 Environmental tests. The following tests shall be performed in accordance with MIL-STD-810. The frequency of operation shall not vary more than 20 percent during any of the following tests.

* 4.4.2.5.1 High temperature test. Each sample shall be subjected to high temperature in accordance with MIL-STD-810. The frequency of operation shall be checked at the beginning of the test, at the completion of 25 hours at high temperature, and at the conclusion of the test. The operating voltage shall be 28 volts.

* 4.4.2.5.2 Low temperature test. Each sample shall be subjected to low temperature in accordance with MIL-STD-810, except the exposure period shall be 72 hours. The frequency of operation shall be checked at the beginning of the test, at the completion of 36 hours at low temperature, and at the conclusion of the test. The operating voltage shall be 28 volts.

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* 4.4.2.5.3 Humidity test. The samples shall be subjected to humidity in accordance with MIL-STD-810. The frequency of operation shall be checked at the beginning and at the conclusion of the test.

* 4.4.2.5.4 Salt spray test. The sample shall be subjected to salt spray for 50 hours in accordance with MIL-STD-810. The frequency of operation shall be checked at the beginning and at the conclusion of the test.

* 4.4.2.5.5 Fungus test. The samples shall be subjected to fungus in accordance with MIL-STD-810.

* 4.4.2.5.6 Sand and dust test. The samples shall be subjected to sand and dust in accordance with MIL-STD-810.

* 4.4.2.5.7 Altitude test. The samples shall be subjected to altitude in accordance with MIL-STD-810 for a period of 15 minutes. During this test, the keyer shall be connected in the circuit as shown on Figure 1 and operated continuously for a period of 5 minutes with an applied voltage of 28 volts DC.

* 4.4.2.6 Vibration test. The samples shall be subjected to vibration in accordance with MIL-STD-810. During this test, the keyer shall be connected in the circuit as shown on Figure 1 and operated continuously during the operation of scanning the frequency range for resonant frequencies. If the unit is of the electrical-mechanical type, then during the balance of the test an ohmmeter shall be used to check if the contacts close. Closing of the contacts, except during the operation of the keyer, shall constitute failure.

* 4.4.2.7 Voltage surges. Each sample shall be subjected to high-voltage spikes in accordance with and as specified in MIL-STD-704.

* 4.4.2.8 Input transient voltage protection test. With 28 volts DC applied to the NO terminal and the minus terminal grounded, there shall be zero voltage at the normally closed (NC) and plus terminals.

* 4.4.2.9 Load rating test. Each sample shall be tested with following loads:

(1) N.C. leg 40MA N.O. leg 1 AMP.

(2) N.C. leg 1 AMP N.O. leg 1 AMP.

* 4.5 Quality conformance tests.

* 4.5.1 Quality conformance tests shall consist of the individual tests (see 4.5.2) and the sampling tests (see 4.5.3).

* 4.5.2 Individual tests. Each keyer shall be subjected to the following tests.

* 4.5.2.1 Examination of product. Each keyer shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

* 4.5.2.2 Each keyer shall be checked in the circuit shown on Figure 1 to determine that it operates.

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* 4.5.2.3 Each unit shipped shall be certified as meeting the Inspection Test Requirements of this specification. Rated lamp load, the flashing rate, starting time and flashing on-off ratio shall conform to design requirements of this specification at applied voltages of 22 volts and 28 volts DC.

* 4.5.3 Sampling tests. Keyers shall be randomly selected from each lot and subjected to the operating frequency tests specified in 4.4.2.1 and 4.4.2.2, the isolation test of 4.4.2.8 and the load test of 4.4.2.9

* 4.5.3.1 Lot. A lot shall consist of keyers manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

* 4.5.3.2 The sample size shall be determined in accordance with MIL-STD-105, General Inspection Level II. The Accepted Quality Level shall be 2.5.

* 4.5.3.3 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes or both, as warranted, and on all units of product which can be corrected and which are manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all tests and examinations, or the test which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

* 4.5.3.4 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of section 5 or the documents specified therein.

* 5. PACKAGING

* 5.1 Preservation. Preservation shall be level A, or commercial as specified (see 6.2).

* 5.1.1 Level A. Warning light keyer shall be adequately cushioned and preserved method 1C-1 of MIL-P-116.

* 5.1.2 Commercial. Warning light keyer shall be preserved in accordance with ASTM D 3951.

* 5.2 Packing. Packing shall be level A, B or commercial as specified (see 6.2).

* 5.2.1 Level A. A quantity of warning light keyers preserved as specified in 5.1 shall be packed in a wood-cleated plywood shipping container conforming to PPP-B-601, overseas type, grade A, style optional. Insofar as practicable, containers shall contain identical quantities and be of minimum cube and tare consistent with the protection required. Closure and strapping shall be in accordance with the appendix to PPP-B-601.

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* 5.2.2 Level B. A quantity of warning light keyers, preserved as specified in 5.1, shall be packed in domestic type, fiberboard shipping container conforming to PPP-B-636, class weather resistant, style if specified. Insofar as practical, containers shall contain identical quantities and be of minimum cube and tare consistent with the protection required. Closure and strapping shall be in accordance with the appendix to the box specification.

* 5.2.3 Commercial. Warning light keyers preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951 as applicable.

* 6. NOTES

* 6.1 Intended use. The type keyer covered by this specification is intended for use in the flickering of a fire warning light to indicate an overheat condition.

* 6.2 Ordering data. Acquisition documents should specify the following:

Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

* 6.3 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 99

Reviewer:
DLA - GS

Preparing activity:
Air Force - 82

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-K-8144B(USAF)	KEYER, OVERHEAT WARNING LIGHT, TYPE MA-1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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